39-year annual time series of dominant tree species and leaf type cover for Italy derived from Landsat time series using deep learning

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Abstract. Consistent long-term maps of forest leaf type cover and dominant tree species at high spatial resolution are currently unavailable for Italy. The study presents annual 30 m resolution maps of continuous evergreen broad-leaved, deciduous, and coniferous leaf type cover (LTC) fractions and dominant tree species (DTS) class probabilities for all forested pixels in Italy spanning 1987–2025, derived from the Landsat Collection-2 Level-2 archive (Landsat 4/5/7/8/9). Time Series Transformer models are trained via a three-stage workflow combining contrastive pretraining with supervised fine-tuning on Italian forest plot observations, forest inventories, and a synthetically generated training dataset to predict both tasks. Pixel-level uncertainty estimates are provided as the standard deviation of a five-seed ensemble for both outputs. A three-year observation pooling strategy is applied to increase cloud-free observation density and mitigate inter-annual variability. Evaluation of the time series maps are based on hold-out datasets across different spatio-temporal scales and a comparison with historical orthophotos. The final products achieve an average MAE of $13.5\,\%_{cover}$ with an R^2 of 0.75 for LTC mapping and an overall accuracy of 0.69 across all species-related DTS classes, with most class confusions occuring within specific leaf types. The 39-year time series enables historical reconstruction of forest composition dynamics and long-term ecological change detection at national scale and relevant time scales for climate-driven species range shifts.

1 Introduction

With over 40 years of continuous acquisitions, the Landsat missions archive contains the longest consistent Earth observations available for large-scale vegetation monitoring (Wulder et al., 2022). Its four decades of continuity makes it uniquely suited for change detection or hindcasting of environmental variables including forest habitat characteristics (Turubanova et al., 2023; Bell et al., 2024). The archive has been applied to discriminate forest types and quantify forest variables such as canopy height or greening trends from regional to continental scales (Pflugmacher et al., 2019; Obuchowicz et al., 2024; Yel et al., 2026). In monitoring long-term changes and disturbances in forest vegetation under climate change, the Landsat archive provides

a valuable modelling base (Viana-Soto and Senf, 2025; Grünig et al., 2026). The slow change and succession dynamics of forest vegetation, which is gradually unfolding over decades, make this long temporal availability particularly valuable. A shorter observation period, e.g. less than 10 years, cannot capture natural stand-level transitions such as species shifts or forest densification (Tong et al., 2023; Midolo et al., 2026). The medium spectral, spatial, and temporal resolution of Landsat (6 reflectance bands, 30 m ground sampling distance, and 8–16 day revisit time) is adequate for capturing annual phenological dynamics at stand level, which demonstrated to be critical for distinguishing tree species and leaf types (Hemmerling et al., 2021; Kollert et al., 2021; Wang et al., 2022).

Despite these strengths, the use of multi-decadal Landsat data for vegetation mapping raises several methodological challenges. The number of usable cloud-free observations has increased since the early periods of the archive, due to the launch of new Landsat missions, which increases temporal resolution. This increase in observation density has been shown to cause non-natural trends in annual maximum Normalized Difference Vegetation Index (NDVI), where up to 50% of observed Landsat-based greening in alpine environments can be attributed to this observational bias rather than true vegetation change (Bayle et al., 2024). A second challenge arises from the failure of the Scan Line Corrector (SLC-off) on Landsat 7 ETM+, which introduces systematic spatial data gaps. Finally, spectral radiometric differences between sensors, which arise from different sensor types, band widths, and data processing procedures across Landsat missions, introduce inter-sensor discrepancies that may produce unintentional changes in modelled variables across the Landsat observation record.

Multi-temporal satellite data consistently outperforms single-date or mosaic-based approaches for tree species and leaf type mapping. Phenologically informative seasons, in particular leaf-off periods and autumn senescence, contribute largely to species discrimination (Kollert et al., 2021; Tan et al., 2025). When sufficient temporal coverage is available, dense time series, that exploit the full phenological cycle, can outperform the inclusion of additional environmental predictor variables such as topography or climate indices (Hemmerling et al., 2021). These findings highlight temporal resolution and coverage as primary sources for improving species-level modelling and mapping performances.

Deep learning, and in particular Transformer-based architectures, provide a framework for processing raw, irregular, multi-band satellite time series (STS). Transformers process sequences of observations without requiring interpolation to a regular sequence, an essential capability for processing long Landsat archives with varying observation density across decades and sensor platforms (Zerveas et al., 2020; Yuan et al., 2022). Self-supervised pretraining on large unlabelled STS or image time series datasets has been shown to improve feature representation quality and performance on downstream tasks, particularly in data-scarce settings (Yuan Yuan et al., 2020; Hiebl et al., 2025; Feng et al., 2026). Contrastive self-supervised learning approaches exploit temporal and seasonal variability by training representations to be invariant across different observation windows of the same location, which is a characteristic suited to stable land cover types such as forest stands (Chen et al., 2020; Mañas et al., 2021). Providing calibrated uncertainty estimates alongside predictions has emerged as an important requirement for ecological mapping applications, to enhance interpretability by users. Deep ensembles of independently trained networks deliver calibrated epistemic uncertainty that is able to identify out-of-distribution inputs and erroneous predictions, when feature and label related domain shifts appear during the mapping phase (Sylvain et al., 2024; Hiebl et al., 2025).

Italian forests are undergoing rapid and spatially complex changes. The Italian National Forest Inventory (INFC2015) documents a continuing expansion of forested area from approximately 10.46 million ha in INFC2005 to 10.98 million ha in INFC2015 driven by land abandonment and reforestation programs (Gasparini et al., 2022). At the same time, climate change in combination with land abandonmnet alters phenological characteristics and forest composition in favor of species with higher drought resilience (Chelli et al., 2017; Herraiz et al., 2025), while thermophilous species are projected to spread into colder forest (Bricca et al., 2026). These processes are spatially heterogeneous, reflecting regional variety in land-use changes, climate change impact, and historical species composition. Despite these dynamics, spatially explicit maps of forest composition and variables in Italy remain restricted to static maps with no continuous temporal resolution. The most recent national reference product, the Carta Forestale d'Italia 2020 (CFI2020), provides a single-epoch forest map at 1:10,000 scale based on photo-interpretation of aerial orthophotos, but does not capture historical trajectories of change nor the full forest extent of Italy (Mattioli et al., 2025).

Our study is filling this gap by providing data-driven, spatio-temporal wall-to-wall maps of leaf type cover (LTC) fractions and dominant tree species (DTS) probabilities for all forested areas in Italy, annually from 1987 to 2025 and at Landsat-native $30\,\mathrm{m}$ resolution. We assemble and process 41-years of Landsat Collection-2 Level-2 time series (Landsat-4/5/7/8/9) and train multiple Time Series Transformer models (TST) via contrastive pretraining and a fine-tuning workflow (Rußwurm and Körner, 2020; Zerveas et al., 2020; Chen et al., 2020). The model leverages per-pixel, irregular Landsat observation time series using Multi-Head Attention and a time encoding. The final product are annual $30m$ maps of i) 11 dominant tree species classes (see. Sec. 2.2.3), and ii) LTC fractions for evergreen broad-leaved (EVE), deciduous broad-leaved (DEC), and coniferous (CON) leaf type. Additional ensemble-based uncertainty estimations are provided per-pixel and target variable to improve interpretability of the produced maps.

2 Materials and Methods

2.1 Study Area

The study covers all forested areas of peninsular and insular Italy, spanning several biogeographical, climatic, and latitudinal regions and the full elevation gradient of the country (Fig. 1). Forest extends from sea level — $34.1\,\%$ of the national forest area lies below $500\,\mathrm{m}$ a.s.l. and $1.1\,\%$ above $2000\,\mathrm{m}$ a.s.l. (Gasparini et al., 2022) — up to the alpine treeline, which in the western Italian Alps was dated at approx. $2500\,\mathrm{m}$ a.s.l. in 2008 (Leonelli et al., 2011). Italian forests fall within alpine, temperate, sub-Mediterranean, and Mediterranean climate zones (Chelli et al., 2017), and are dominated by four forest habitat types: broadleaved-deciduous and temperate coniferous forests concentrated in the Alps and Apennines, and evergreen broadleaved and Mediterranean coniferous forests concentrated along the coasts and in the south (Bricca et al., 2026). This climatic and ecological heterogeneity is a central challenge for seamless forest mapping, as models trained predominantly on well-sampled regions risk out-of-domain predictions in climatically and ecologically different areas with sparser reference data. Forests cover 37% of Italy's territory (10.98 million ha) (Gasparini et al., 2022). Italy's position as a transition zone between the Mediterranean and the temperate climate of Central Europe makes its forests especially vulnerable to climate change,

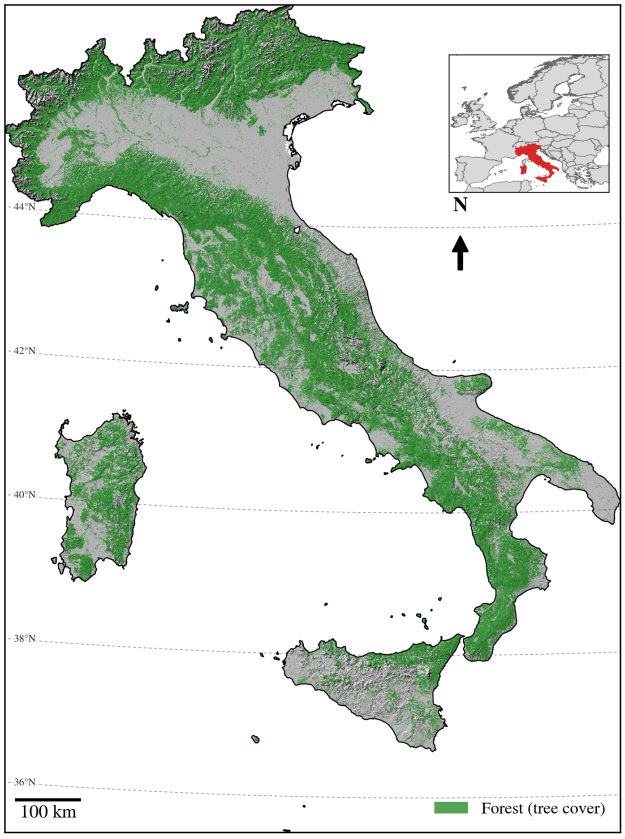


Figure 1. Study Area Italy with high forest cover in the Alps and mountainous areas of Central Italy. Forested areas in green are depicted according to the ESA WorldCover map (Zanaga et al., 2022).

as warming and increasing drought frequency are already reported to shift timing of leaf-out and senescence phenology and favour more drought-resilient and evergreen broad-leaved species often at the cost of deciduous ones (Chelli et al., 2017; Herraiz et al., 2025). These climate-driven shifts compound ongoing changes in forest management and land use, reshaping tree species distribution and forest biodiversity across the country (Bricca et al., 2026).

2.2 Modelling Data

Six different datasets are used for training and validation of the models, with two field campaign datasets covering DTS and LTC, two remotely interpreted dataset for DTS, and two synthetic datasets for LTC. An overview of the used datasets during training are provided in Tab. 1 and Fig. 2.

Table 1. Overview of datasets used during model training and validation for Leaf Type Cover (LTC) and Dominant Tree Species (DTS) mapping.

	n samples	LTC	DTS	Characteristics
VPO	2016	yes	yes	conducted in the field 2023-2025, standardized $10\,\mathrm{m} \times 10\,\mathrm{m}$ plots (see Sec. 2.2.2)
VDBI	2998	yes	yes	observations with varying size, acquisition time, and location accuracy (see Sec. 2.2.1)
rsCFI	131095	-	yes	randomly sampled across CFI dominant tree species (see Sec. 2.2.3)
csCFI	30380	-	yes	pure stand polygons sampled from CFI (see Sec. 2.2.3)
rsCOVER	131095	yes	-	same locations as rsCFI, synthetically created using AlphaEarth embeddings (see Sec. 2.2.4)
csCOVER	30380	yes	-	same locations as csCFI, synthetically created using AlphaEarth embeddings (see Sec. 2.2.4)

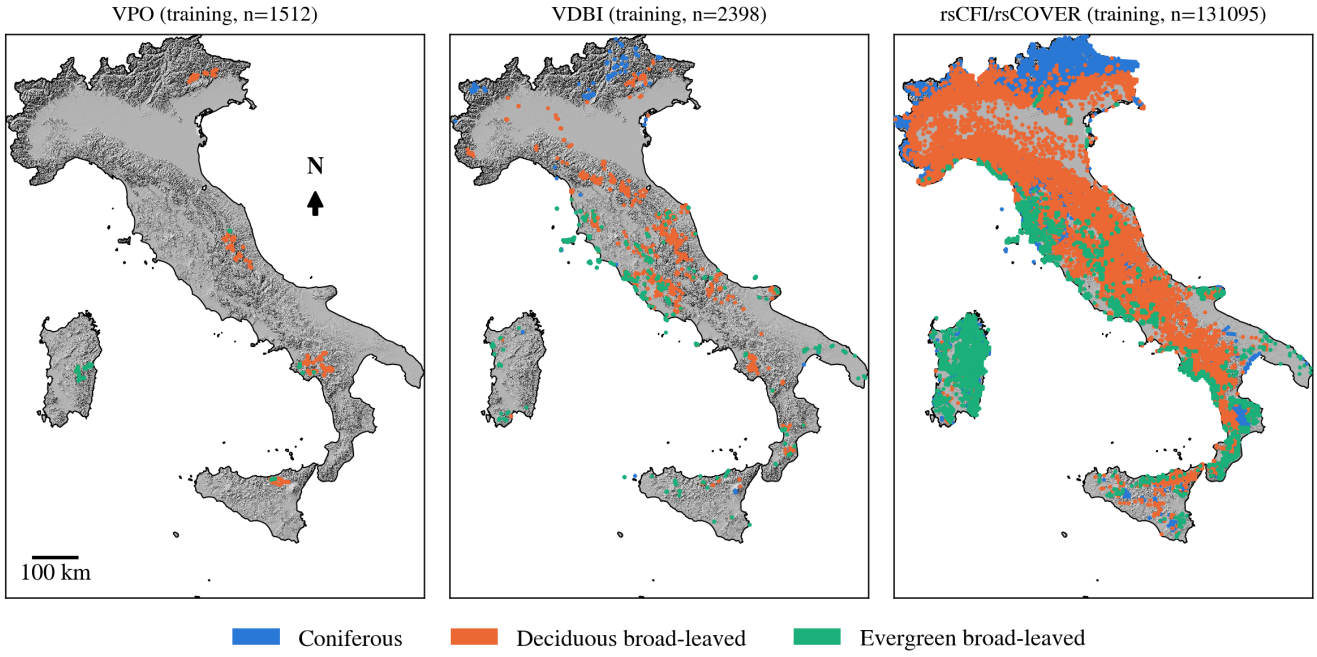


Figure 2. Overview of training datasets used in the study.

2.2.1 Italian forest vegetation plot databases

The Forest Vegetation Database Italy (VDBI) is a compilation of over 16,000 phytosociological and vegetation relevé plot observations collected across Italy from the 1970s to the present. Each plot record contains species composition data, structural attributes, and cover estimates for the tree layer (Alessi et al., 2019). The VDBI covers the full range of Italian forest types from Mediterranean habitats to subalpine conifer belts. Temporally it provides data back to approximately 1972, enabling historical

training data for pre-2000 Landsat time series. The location accuracy of single plot observations in the dataset varies across observation year and source dataset from $< 10\,\mathrm{m}$ to $1000\,\mathrm{m}$, which is a significant source of spatial bias that challenges use as validation data (see also Sec. 2.6). To avoid non-forest pixels leak into the training and test datasets, VDBI was cleaned manually based on orthophoto imagery. The final dataset after the cleaning process provides 9837 plot observations. Plot records include LTC estimates (EVE, DEC, CON) and tree species cover fractions, which were used to retrieve dominant DTS classification labels, allowing training of the regression ($VDBI_{LTC}$) and classification ($VDBI_{DTS}$) TST model. A $10\,\mathrm{km} \times 10\,\mathrm{km}$ regular polygon grid was created for the train/validation/test split to reduce spatial autocorrelation between training and validation sets. A 20% validation split, stratified by polygon identifier is applied throughout all training stages (Safonova et al., 2023).

2.2.2 Field plot observations

The Vegetation Plot Observations (VPO) dataset consists of recent field campaigns conducted within 7 Italian National Parks or protected areas (Hiebl et al., 2025, 2026b). Each site comprises four systematic $10\,\mathrm{m} \times 10\,\mathrm{m}$ vegetation plots arranged on a regular grid, within which species-wise cover is estimated for three vertical layers ($0 - 1\,\mathrm{m}$, $1 - 10\,\mathrm{m}$, and above $10\,\mathrm{m}$). LTC is extracted as sum over verical layers of EVE, DEC, and CON species cover fractions, while DTS is - similar to VDBI - retrieved from species cover by selecting the tree species with highest cover fraction. The VPO data used in this study consists of observations from 2023 to 2025 field season, yielding a total of 2016 individual plot observations from 504 sites. The VPO dataset provides high-quality, spatially clustered reference data for the most recent Landsat time steps (2023-2025), and serves as the primary validation reference for model performance in the recent high-density observation period. The VPO dataset is further split into a training set (80%) and a test set (20%) to evaluate model performance. The split was performed spatially after clustering the sites by their coordinates into 10 clusters per region using a constrained k-means algorithm (Bradley et al., 2000). The spatial clustering was done to stabilize training and prevent over-optimistic accuracy estimates due to spatial autocorrelation (Safonova et al., 2023; Hiebl et al., 2025).

2.2.3 Carta Forestale d'Italia - Forest Inventory Data

The *Carta Forestale d'Italia 2020* (CFI2020) is the first national forest map of Italy produced at operational scale (1:10,000), based on photointerpretation of aerial orthophotos from 2018 to 2020 against different forest definitions (Mattioli et al., 2025). The CFI2020 delineates forested areas across all Italian regions and provides stand-level attributes including forest category, silvicultural system, and DTS group. The initial 43 unique classes in the dataset were collapsed into 11 dominant tree species classes: silver fir (*Abies alba* Mill.), spruce (*Picea abies*), pine (*Pinus* sp.), deciduous oak (deciduous *Quercus* sp.), chestnut (*Castanea sativa*), beech (*Fagus sylvatica*), manna ash - hornbeam (/textitornus–ostrya, *Fraxinus ornus–Ostrya carpinifolia*), evergreen oak (*Quercus ilex/suber*), coniferous others, deciduous others, and evergreen others. In this study, the CFI2020 polygon layer serves two roles: i) as an additional training/validation data source for the DTS classification task, and ii) as a source of decision rules for cleaning the synthetic LTC training dataset (see Sec. 2.2.4). In the first case two different datasets are created by i) randomly sampling a set of evenly distributed points over all classes (rsCFI), and ii) by sampling points from

manually delineated pure stand polygons that are evenly distributed across classes (csCFI). In the second case, CFI2020 species attribution (DTS) is used to impose ecologically plausible constraints on predicted cover fractions of the synthetic dataset (see Sec. 2.2.4).

2.2.4 Synthetic leaf type cover data

Spatially explicit seamless LTC data for model training are not available for Italy. Roughly following the approach of Kang et al. (2021) and Tan et al. (2025), who demonstrate the use of coarser or indirectly derived data sources as proxy training targets where direct measurements are absent, an artificial LTC dataset was created from the VDBI and VPO plot observations. A Random Forest regression model was trained on the combined VDBI and VPO training split, with EVE, DEC, and CON cover fractions as prediction targets. As input features, AlphaEarth embeddings (Brown et al., 2025; Hiebl et al., 2026b), which are multi-modal spectral-spatial-temporal feature representations derived from a globally pretrained geospatial foundation model, were extracted for all training plot locations. AlphaEarth embeddings were shown to achieve high species classification accuracy with very limited training data in Italian mountain forests (Ball et al., 2026), making them particularly suited as inputs for the pseudo-label generation step.

The trained Random Forest model was then applied to the full CFI2020 forest extent to generate a nation-wide pixel-level pseudo-label layer for each cover fraction. The labels were extracted at the same locations as rsCFI (rsCOVER)and csCFI (csCOVER). This pseudo-label layer constitutes the primary continuous-cover target during the contrastive pretraining and early fine-tuning stages. To reduce label noise introduced by model prediction errors, the CFI2020 species attribution was used to apply ecologically plausible cleaning rules: e.g. pixels within pine-dominated forest polygons were required to have at least 60% CON cover and fewer than 30% EVE cover; analogous rules were applied to classes of beech, oak, and other dominant tree species. With these cleaning rules parts of the pseudo-labelling errors are removed, while preserving the spatial completeness of the training dataset. If a sampled synthetic plot does not meet the criteria the sample is dropped. This way we prevent ecologically non-feasible labels from being forwarded into the final training loop.

2.2.5 Landsat time series

Annual Landsat time series were fetched for the full Italian national territory from Microsoft Planetary Computers STAC Catalog using Landsat Collection-2 Level-2 surface reflectance products from Landsat 4 TM (1982–1993), Landsat 5 TM (1984–2013), Landsat 7 ETM+ (1999–2023), Landsat 8 OLI (2013–present), and Landsat 9 OLI-2 (2021–present) (EROS, 2022a; EROS, 2022b; EROS, 2022c). Scenes with cloud cover higher than 75% were excluded, which led to a per year count of approx. 80 to 120 scenes. Per-pixel cloud, cloud shadow, cirrus, and snow masking was performed using the Landsat Collection-2 Level-2 QA PIXEL quality bitfield. For each forested pixel, 14 spectral features are computed at each valid observation: six surface reflectance bands (blue, green, red, near infrared, SWIR1, SWIR2) and eight derived spectral indices (NDVI, NIRv, NDMI, EVI, WDRVI, NBR, NDWI, GNDVI) (Montero et al., 2023). For mapping, the Landsat scenes were fetched and preprocessed on-the-fly in $25\,\mathrm{km} \times 25\,\mathrm{km}$ chunks from Microsoft Planetary Computer parallel to the GPU-based model predictions. The number of valid observations per year varies across the 40 year archive depending on the missions

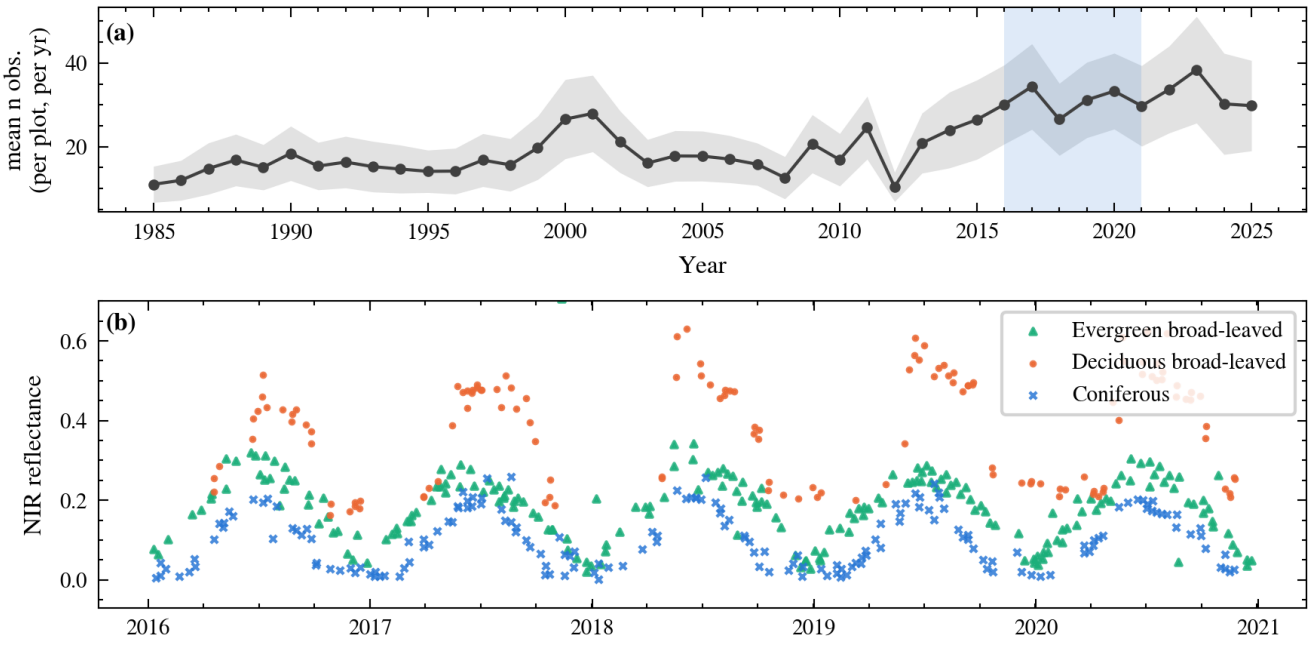


Figure 3. a) Number of valid observations per year averaged over all plot observations in the csCFI dataset. b) Examples of Landsat time series for evergreen oak, beech, and spruce dominated plots for a 5 year time period, highlighting the irregularity of observations, inter-annual variability, and differences in phenological patterns between species.

overlap and generally increases over time. Annual phenology patterns are used to differentiate between species and cover types in the models (Fig. 3).

2.2.6 Additional data sources

The ESA WorldCover product is used as an initial broad-scale delineation of forested areas across Italy (Zanaga et al., 2022). Only pixels labelled as tree cover are forwarded for prediction in the final maps. We are aware that the ESA WorldCover product does not provide a sufficient forest mask for detailed analysis of the datasets, as many mixed pixels and edge cases are remaining. But by applying a conservative masking during mapping we leave room for stricter forest delineation through other products without loosing regional accuracy.

2.3 Time Series Transformer Architecture

The backbone model is an encoder-only Time Series Transformer (TST) extended to support irregular, sparse, annual satellite image time series. The architecture employs multi-head self-attention across the temporal dimension, enabling the model to learn both short-range phenological features and long-range inter-seasonal dependencies within the aggregated 3-year observation window (Rußwurm and Körner, 2020; Zerveas et al., 2020).

For each target year, three consecutive years of Landsat observations are concatenated, encompassing the target year and the two immediately preceding years (year-2, year-1, year), and aggregated into a single annual cycle to densify the sparse annual time series. After aggregation, observations are chronologically sorted and padded to a fixed sequence length of 200 time steps. Positions corresponding to padded time steps receive a key-padding mask that prevents these positions from contributing to attention score computations (Vaswani et al., 2023). The actual acquisition day-of-year for each observation is encoded as a sinusoidal positional encoding, enabling the model to correctly reconstruct the irregular temporal spacing of Landsat acquisitions and to distinguish phenological phases across the observation window without requiring interpolation to a regular grid (Yuan Yuan et al., 2020). Input features are normalised per band using a robust standardisation based on the median and interquartile range (quantiles q=0.1 and q=0.9) — with transform parameters fitted on the training data and frozen at inference time to prevent data leakage. A sensor platform embedding with $sensordim = 5$ encodes a multi-hot vector per year indicating which Landsat satellites (L4, L5, L7, L8, L9) contributed to the observations. This embedding is added to the pre-attention temporal encoding via a learned linear projection and makes the Transformer aware of inter-sensor radiometric and observation density differences over the 40-year archive. A post-attention pooling layer using a learned query vector collapses the temporal dimension into one single summary vector per output target, letting each targets learned query decide, which valid timesteps matter most for its prediction. In a setting with target classes from a wide range of phenological and spectral characteristics, this pooling strategy allows the model to learn class-specific temporal patterns across the observation window. The model heads are two-layer Multi-Layer-Perceptrons (Hendrycks and Gimpel, 2016).

The classification model version maps the Transformer's output to probabilities over the eleven DTS classes and three leaf type classes trained with two separate Cross Entropy Loss (CELoss) computations summed into a single loss value (dominant species and leaf type). With the dual loss the model learns to simultaneously capture species identity without losing the ability to capture the distinction between EVE and CON leaf types. The regression model version are trained with Mean Squared Error Loss (MSELoss) and maps the output to continuous EVE, DEC, and CON cover fractions (LTC).

2.4 Training procedure

The model is trained via a three-stage progressive workflow designed to maximise temporal stability and to accommodate the heterogeneous quality and temporal coverage of available reference data (Fig. 4).

Stage 1 - Contrastive pretraining. The backbone is trained using a combined loss that simultaneously optimises a supervised objective (DTS or LTC) using CELoss/MSELoss and an unsupervised normalised temperature-scaled cross-entropy contrastive loss (NT-Xent) (Chen et al., 2020): $L = \lambda \cdot$ NT-Xent + CELoss. For the contrastive term, positive pairs are formed from the same forest plot observed in two independently sampled three-year windows ($year_0$ and $year_1$ drawn at random), representing different but ecologically equivalent observation contexts for the same location. NT-Xent pushes the backbone representations of these positive pairs toward each other in embedding space while repelling representations of observations from different plots. This objective forces the backbone to learn year-invariant spectral-temporal features that are not distracted by observation density or inter-annual radiometric variability (Mañas et al., 2021). Further diversification of positive pairs was done using time series augmentation, which applies three independent changes to each view: random temporal masking (up to 20% of

valid time steps zeroed), spectral noise injection (additive Gaussian noise with $\sigma = 0.01$), and band masking (10% of spectral bands zeroed per sample) (Yuan et al., 2025). For contrastive pretraining only data from csCFI and csCOVER are used, as we consider these pure forest stands as stable over time. The contrastive batch and supervised batch are routed through separate heads attached to the shared backbone: a two-layer MLP projection head for NT-Xent, and the classification/regression head for CELoss or MSELoss, respectively. The model is optimised using AdamW with a cosine learning rate schedule and a 10% warm-up phase. Class imbalance is addressed by class-weighted CELoss and by sample weights set to the square root of the inverse class frequency, following the approach of Grabska-Szwagrzyk et al. (2024). A stratified 20% validation hold-out, stratified by plot polygon identifier, is used across all training stages to prevent spatial autocorrelation between training and validation sets.

Stage 2 - Fine-tuning on labelled data. The contrastive backbone from Stage 1 is fine-tuned using a purely supervised objective (CELoss and MSELoss). The embedding layer is kept frozen in this stage to preserve the contrastive representations, while the task heads are retrained using a reduced learning rate to prevent catastrophic forgetting (Hiebl et al., 2025). This stage runs on the full 1985–2025 Landsat time series, allowing the model to adapt the task heads to the full historical archive without disturbing the temporally invariant representations established during contrastive pretraining. The used reference data are the VDBI historical plot observations.

Stage 3 - Fine-tuning on plot observation data. Additional fine-tuning passes are applied from the Stage 2 checkpoint giving priority to recent high-quality VPO field observations that provide the most accurate ground reference data for the present forest state. This stage is particularly intended to base the model predictions of recent (post-2020) DTS and LTC to field-validated reference values, while the contrastive representations established in earlier stages ensure that the model remains consistent with historical predictions.

Five independent model instances with different random seeds, architectural choices within TST, and random subsets of training and validation data, are trained per stage. The five-seed ensemble delivers two outputs per pixel per year: the ensemble mean prediction (used as the primary data product) and the inter-seed standard deviation (used as the pixel-level uncertainty estimate). High inter-ensemble uncertainty identifies pixels, where the model predictions are unstable, typically in regions with low observation density, unusual spectral characteristics, or high class confusion in mixed stands.

2.5 Mapping scheme

The forest prediction region is defined by the ESA WorldCover forested area (see 2.2.6). For each target year from 1985 to 2025, the corresponding three-year Landsat observation window (year-2 to year) is fetched from the Landsat Collection-2 Level-2 archive, cloud-masked, and processed to the 14 spectral features described in Section 2.2.5. Features are normalised with the fixed training-time standardization parameters (Sec. 2.3) and are padded to the fixed 200 time steps of the TST input sequence. The five trained ensemble models are applied independently to each forested pixel, and the resulting class probability distributions and cover fraction predictions are aggregated across ensemble members to produce the ensemble mean prediction and inter-ensemble uncertainty layers. The uncertainty layers are reported as the standard deviation of the ensemble predictions

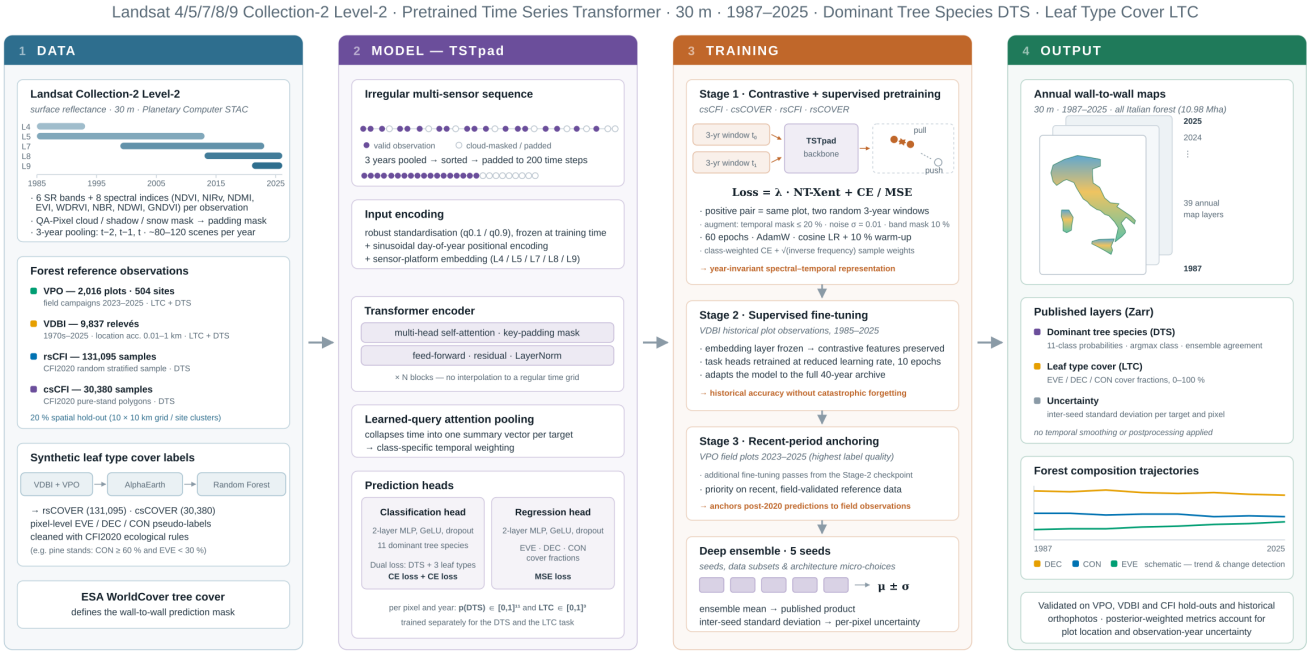


Figure 4. Workflow diagram illustrating the three-stage progressive training approach for forest mapping using Landsat time series data.

for each LTC fraction and DTS class probability (DTS). For DTS maps a level-of-agreement (LOA) between the ensemble members for the majority class is additionally reported as measure of uncertainty for the argmax class.

Observation density varies substantially across the archive. Before 1999, when only Landsat 5 TM was operational, the three-year window typically yields fewer cloud-free observations than the post-2013 period when Landsat 8 and 9 operate simultaneously (Fig. 3). This density gradient is the observational sampling bias described for NDVI trend analyses by Bayle et al. (2024). Predictions in the early archive period carry higher uncertainty due to sparse temporal sampling, which may cause the model to have lower confidence.

We did not process the resulting annual time series using outlier detection or temporal smoothing, before storing the published LTC and DTS prediction maps. This was a design choice so downstream users can apply temporal smoothing or outlier detection methods specifically appropriate to their application context. This is a necessary step for downstream tasks, as the time series contain both true ecological change, but also unintentional noise from observation gaps, cloud contamination, and inter-annual spectral variability.

2.6 Evaluation

The validation strategy is designed to assess both the accuracy of the map product at recent time steps and the temporal consistency of predictions across the full 40-year archive. Three independent validation datasets are used. The VDBI hold-out test split provides the primary historical validation reference, with plot observations dating back to approximately 1995. This

split allows accuracy assessment at historical time steps (1995–2025) where field data are available, spanning the Landsat 5 and Landsat 5/7 overlap periods. The VPO hold-out split — collected during the 2023 to 2025 field work — provides the primary validation reference for recent predictions (2023-2025) from the high observations density Landsat 8/9 period. This dataset contains data from many mixed forest stands with hard-to-predict labels. A third validation source consists of a set of pure-stand forest polygons manually interpreted from the CFI dataset in combination with high-resolution orthophoto imagery at recent time steps. These polygons, selected to represent spectrally explicit mono-species stands, enable validation of the classification output under conditions where the expected DTS label is reliable, unconstrained by label uncertainty in mixed stands or uncertain plot observation coordinates.

For the classification head, overall accuracy (OA) and per-class F1 score are computed for each period. For the regression heads, root mean squared error (RMSE) and mean absolute error (MAE) are reported for each of the three cover fractions (EVE, DEC, CON). The VPO dataset is finally used to assess in detail the model's ability to predict DTS and LTC in mixed stands, where the prediction of a single label is more difficult and uncertain.

Additionally historical ortophoto imagery from several regions in Italy and different years are used to visually assess the temporal consistency of the predicted maps with respect to observed changes in forest cover.

To account for uncertainties in plot observations coming from (i) a spatial mismatch between plot location and Landsat pixel, and (ii) temporal uncertainty and variations between plot observation year and Landsat acquisition year, all performance metrics are computed under three estimators (Appendix A). The *direct* estimator reads the map at the pixel containing the reported plot coordinate in the reported observation year and uses no information about the reference label. The *posterior-weighted* estimator averages the map over a space–time neighbourhood of the plot, weighting each candidate pixel by a Gaussian prior over the reported location accuracy and observation year, combined with a Gaussian likelihood that downweights candidates whose predicted value disagrees with the reference label, following the likelihood-based geolocation correction method (Lobo et al., 2020; Shannon et al., 2024). The *best-match* estimator is the limiting case of the posterior-weighted estimator for likelihood bandwidth $\sigma_{\mathrm{obs},k} \to 0$, and selects the single candidate that agrees most closely with the reference label within the location accuracy neighborhood. The posterior-weighted estimator is bracketed by the direct estimator that ignores the label entirely and is therefore pessimistic, and the best-match estimate that is overly optimistically biased. We report performance metrics (RMSE, MAE and R^2; Precision, Recall, and F1-Score) computed under all three methods to characterize how sensitive model accuracy is to possible location and temporal errors (see Appendix A).

3 Results and Discussion

The final maps provide following variables per pixel and per year (1987 to 2025):

- Dominant tree species: ensemble mean of dominant tree species class and level-of-agreement across ensemble members; ensemble mean of class probabilities and ensemble standard deviation for 11 classes
- Leaf type cover fractions: ensemble mean of EVE, DEC, and CON cover fractions and ensemble standard deviation

3.1 Performance metrics on test datasets

3.1.1 Leaf type cover regression

The hold-out splits of the recent VPO dataset and of the multi-temporal VDBI collection are used to quantify the error of the annual LTC maps (Tab. 2). VPO plots were located with an accuracy of approximately $5\,\mathrm{m} - 10\,\mathrm{m}$, which is high relative to the $30\,\mathrm{m}$ Landsat grid, so the estimator evaluated directly at the reported plot coordinate is the appropriate method for this dataset, while for VDBI, the combination of the three estimators can give insights into the sensitivity of the metrics to residual location and timing error (Sec. 2.6; see also Appendix A).

Evaluated directly at plot location, the maps reproduce DEC cover with an MAE of $9.7\,\%_{cover}$ ($R^2 = 0.84$) and EVE cover with an MAE of $12.5\,\%_{cover}$ ($R^2 = 0.66$; Tab. 2). The consistently weaker performance for EVE has two plausible explanations. First, the RMSE–MAE gap is larger for EVE than for DEC, indicating that the EVE error is driven by a comparatively small number of large residuals rather than by a uniform offset, i.e. by a subset of stands that is systematically misinterpreted rather than by generally erroneous predictions. Second, VPO reference cover is recorded separately for three vertical strata and aggregated across them, whereas Landsat observes a top-of-canopy signal. Parts of Italian EVE species frequently occur as understory vegetation beneath a deciduous overstory (Conedera et al., 2018; Alessi et al., 2019), a setting that is only weakly reflected in canopy spectral signal outside the leaf-off window (Chastain and Townsend, 2007; Yang et al., 2023). Reference EVE cover can therefore exceed the fraction of the signal that is physically available to the sensor, which inflates the model error.

CON cover has the lowest absolute errors on the VPO split (MAE $5.0\,\%_{cover}$, RMSE $9.1\,\%_{cover}$) but a negative coefficient of determination ($R^2 = -0.05$). This combination is a characteristic of the samples in the dataset rather than the models predictions. The VPO field data contain almost no CON-dominated stands, which means the CON variance is very small as all references and predictions are close to $0.0\,\%_{cover}$.

On the VDBI split the errors increase substantially for all three LTC fractions, with MAE and RMSE rising by a factor of 1.6 to 1.9 relative to VPO and explained variance dropping accordingly for EVE and DEC (Tab. 2). CON is the only fraction that improves on VDBI, which confirms the interpretation given above: VDBI covers the alpine and Apennine conifer belts and therefore contains the reference variance that the VPO dataset lacks, so R^2 is an informative statistic for this fraction only on VDBI. This dataset aggregates plots collected over five decades by different observers, many of them without or with low-accuracy GNSS positioning, with reported location accuracies ranging from 0.01 to $1\,\mathrm{km}$ (Sec. 2.2.1). Positioning errors of this magnitude propagate directly into the accuracy metrics, as field plot positioning error is known to bias remote sensing accuracy assessments negatively (Persson et al., 2022). Under closed conifer canopy, GNSS positioning commonly deviates by several metres to tens of metres (Strunk et al., 2025), so a reported VDBI coordinate may address a Landsat pixel that is spectrally unrelated to the described stand. The VDBI figures evaluated directly at the reported coordinate are consequently not a clean estimate of model error, and we report them as a conservative lower bound on historical map quality rather than as an accuracy statement. Under the posterior-weighted estimator, which accounts for the location accuracy reported for each individual plot, the VDBI errors fall back close to the VPO figures, which gives a more optimistic estimate of the model error.

Table 2. Regression performance (MAE, RMSE, R^2) for leaf type cover (LTC) prediction, evaluated on two test datasets. Each cell reports the three estimators as direct / posterior-weighted / best-match (Sec. 2.6, Appendix A). Bold marks the estimator taken as the reference value for the respective dataset: direct for VPO, posterior-weighted for VDBI. MAE and RMSE are reported as $\%_{cover}$.

	VPO test split			VDBI test split		
target	MAE	RMSE	R^2	MAE	RMSE	R^2
EVE	**12.5** / 11.6 / 8.5	**19.1** / 17.5 / 14.9	**0.66** / 0.72 / 0.80	19.9 / **17.3** /10.0	29.5 / **25.1** / 19.8	0.38 / **0.55** / 0.72
DEC	**9.7** / 8.7 / 8.4	**12.4** / 10.6 / 7.9	**0.84** / 0.85 / 0.92	18.9 / **12.6** / 6.6	25.3 / **16.0** / 11.6	0.51 / **0.82** / 0.90
CON	5.0 / 4.6 / 3.0	9.1 / 8.3 / 6.1	-0.05 / 0.31 / 0.54	8.7 / **6.0** / 3.1	16.2 / **10.4** / 8.1	0.68 / **0.89** / 0.93

The three estimator variants bound this effect quantitatively. Moving from the direct estimator through the posterior-weighted to the best-match candidate lowers MAE and raises R^2 for every fraction on both datasets (Tab. 2), and the width of this bracket differs notably between the two datasets, changing proportionally with their reported coordinate accuracy. It is narrow on VPO and roughly an order of magnitude wider on VDBI. This is what would be expected if reference geolocation, rather than model error, dominated the VDBI metrics.

3.1.2 Dominant tree species classification

DTS classification performance is evaluated on the VPO, VDBI, and CFI test datasets (Tab. 3). VPO and CFI metrics are reported under the direct estimator and VDBI metrics under the posterior-weighted estimator. The model reaches a weighted-average F1-score of 0.62 on the VPO split (macro F1 0.65 over the six classes represented), 0.66 on the VDBI split (macro F1 0.63 over eight classes), and 0.72 on the CFI test dataset (macro F1 0.71 over all eleven classes). Overall accuracy is 0.62 and 0.64 on the plot-based splits (VPO, $n = 477$; VDBI, $n = 605$) and 0.71 on the considerably larger CFI test dataset ($n = 9202$). The gap of $7 - 9\,\%$ between the plot-based splits and the polygon-based CFI evaluation reflects the differing label quality, plot location accuracy but also the mixed stand character of VPO and VDBI datasets compared to CFI dataset.

The per-class results are asymmetric between precision and recall. On the VPO split beech is predicted with high recall but low precision (0.91 / 0.57), whereas deciduous oak shows the inverse pattern (0.46 / 0.88). The model assigns beech to a substantial share of oak-dominated plots, but the predicted beech plots are mostly correct. But the direction of this asymmetry is not stable across the three evaluations. On VDBI and CFI beech behaves in the opposite way and is the more conservatively assigned class (precision 0.89 and 0.78 at recall 0.82 and 0.64), so the over-prediction of beech is a property of the VPO plot sample, dominated by mixed Apennine stands, and not a general property of the model. This highlights the discrepancy of labelling in different datasets. What is common to all three evaluations is the confusion of spectrally and phenologically similar temperate broad-leaved species, which is also documented other national-scale Sentinel-2 mapping (Hemmerling et al., 2021; Grabska-Szwagrzyk et al., 2024). The manna ash–hornbeam class (ornus–ostrya) exhibits the lowest accuracy metrics of all species-level classes across all three datasets (F1 0.45–0.63). These stands are typically structurally heterogeneous and

Table 3. Classification performance (precision, recall, f1-score) for the dominant tree species classification task, evaluated on three test datasets. VDBI metrics are computed with the posterior-weighted estimator based on the location accuracy of the plots, VPO and CFI metrics are computed using the direct estimator (Sec. 2.6, Appendix A1).

	VPO test split				VDBI test split				CFI test dataset			
class	prec.	recall	f1-score	support	prec.	recall	f1-score	support	pre.	recall	f1-score	support
silver fir	-	-	-	-	-	-	-	-	0.84	0.89	0.86	827
other dec.	0.52	0.55	0.53	112	0.31	0.71	0.45	36	0.35	0.79	0.49	276
dec. oak	0.88	0.46	0.60	127	0.71	0.45	0.53	131	0.68	0.64	0.66	1939
chestnut	0.96	0.88	0.92	26	0.80	0.57	0.67	28	0.67	0.66	0.66	962
other con.	-	-	-	-	-	-	-	-	0.46	0.63	0.53	390
beech	0.57	0.91	0.68	108	0.89	0.82	0.85	136	0.78	0.64	0.70	770
eve. oak	0.65	0.64	0.65	53	0.67	0.71	0.70	97	0.86	0.75	0.80	1435
other eve.	-	-	-	-	-	-	-	-	0.89	0.81	0.85	503
pine	-	-	-	-	0.94	0.67	0.78	98	0.71	0.70	0.70	435
ornus-ostr.	0.69	0.43	0.53	51	0.41	0.50	0.45	64	0.61	0.66	0.63	952
spruce	-	-	-	-	0.78	0.47	0.58	15	0.96	0.81	0.88	713
accuracy			0.62	477			0.64	605			0.71	9202
macro avg	0.71	0.65	0.65		0.69	0.61	0.63		0.71	0.73	0.71	
weighted avg	0.68	0.62	0.62		0.73	0.64	0.66		0.74	0.71	0.72	

successional, so a single dominant-species label is a bad characterisation of the pixel. The aggregate "other deciduous", "other coniferous" and "other evergreen" classes behave as expected for categories, that are supposed to catch residual classes. On the VDBI split, "other deciduous" reaches a recall of 0.71 at a precision of 0.31, which means it absorbs plots that the model cannot attribute to any specific class.

Per-class metrics based on small test dataste size require caution. The highest F1-score in the VPO evaluation (chestnut, 0.92) rests on 26 test plots, and the spruce metrics in the VDBI evaluation on 15 plots.The challenges of estimating reliable per-class accuracies for infrequent classes under high class imbalance is a common limitation of large-area species mapping (Hemmerling et al., 2021; Grabska-Szwagrzyk et al., 2024). Five of the eleven classes (silver fir, spruce, pine, other coniferous, and other evergreen) are not represented in the VPO split at all, because the 2023–2025 campaigns targeted specific protected areas across central and southern Italy. VPO therefore constrains the recent-period accuracy of the broad-leaved classes only, and the alpine conifer classes are evaluated exclusively through VDBI and CFI.

The confusion matrix for the CFI test dataset (Fig. 5) shows which class pairs carry the residual error. Misclassification mostly stays within the leaf type of the underlying class. Of the 2640 misclassified plots, 1731 ($66\,\%$) are assigned to another class of the same leaf type group, so the correct leaf type is predicted for $90.1\,\%$ of all test plots while the species label is

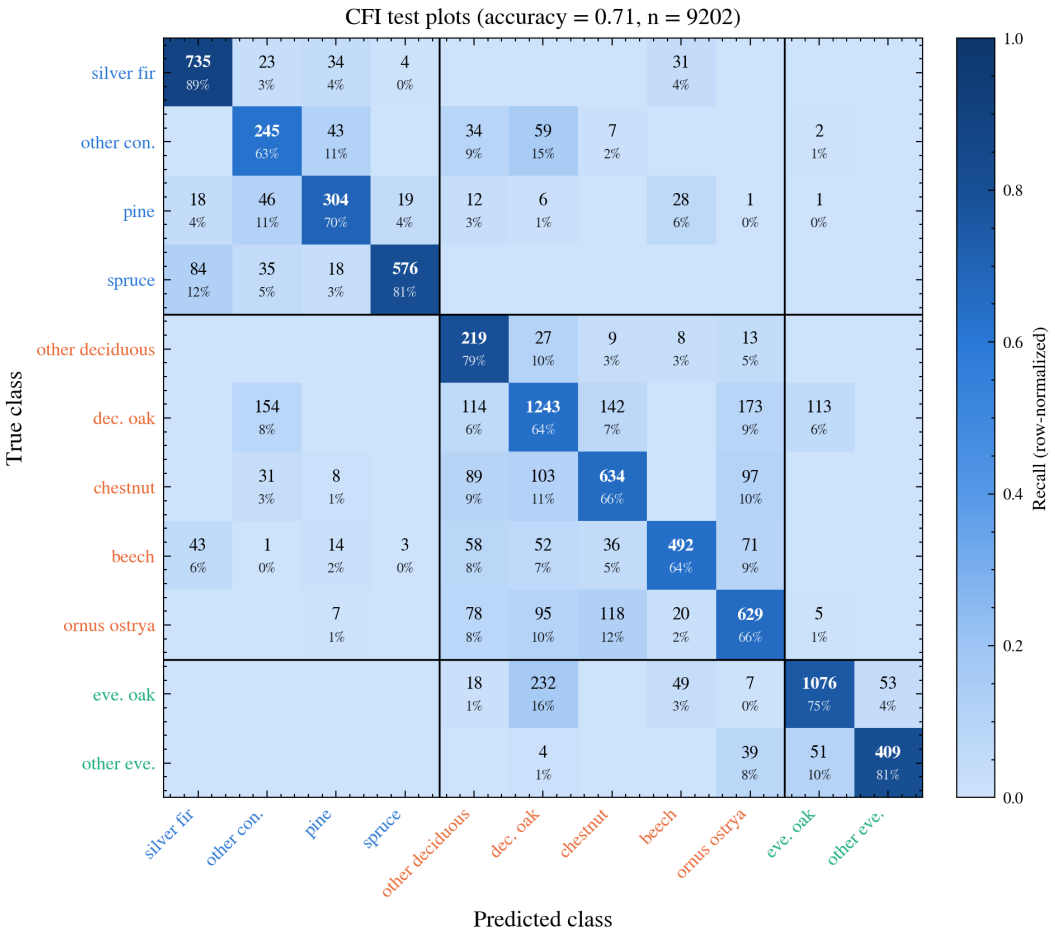


Figure 5. Row-normalised confusion matrix (recall) for the dominant tree species classification on the CFI test dataset ($n = 9202$), computed using the direct estimator (Sec. 2.6). Cell labels give the absolute number of test plots and the row-normalised share. Classes are ordered and grouped by leaf type (coniferous, deciduous broad-leaved, evergreen broad-leaved); black lines delineate the three groups, so that off-diagonal cells outside the diagonal blocks are confusions that also change the predicted leaf type.

correct for only 71 %. The DTS layer is therefore substantially more reliable when aggregated to leaf type than when read at species level, which is consistent with the LTC product being the more robust of the two data layers.

The weakest classes are the two aggregate categories “other deciduous” (F1 0.49) and “other coniferous” (F1 0.53, recall 0.63, precision 0.46). Of the “other coniferous” plots, 11 % are predicted as pine, but 24 % are assigned to “other deciduous” and to deciduous oak. Together with evergreen oak, it is one of only two classes in which the majority of the error crosses the leaf type boundary, indicating that the “other coniferous” category is heterogeneous and does not have a consistent spectral–temporal signature. Within the deciduous classes, deciduous oak has the lowest recall (0.64, same as beech) despite the largest support ($n = 1939$), with the error spread mostly across manna ash–hornbeam (9 %), “other coniferous” (8 %), chestnut (7 %),

and evergreen oak (6 %). The reverse direction is equally strong, with 12 % of manna ash–hornbeam plots being predicted as chestnut and 16 % of evergreen oak plots predicted as deciduous oak. Deciduous oak, chestnut and manna ash–hornbeam therefore form three class pairs, that get confused frequently. The three are co.-occuring across the sub-Mediterranean and lower montane belt and frequently form mixed stands in which a single dominant-species label cannot be described by a 30 m pixel (Bricca et al., 2026).

The deciduous oak–evergreen oak confusion (113 and 232 plots in the two directions) is the largest confusion in the matrix, that occurs across leaf types. It affects the transition that is of primary ecological interest, as these are stands in which evergreen and deciduous oaks co-occur and in which evergreen broad-leaved species happen to expand (Conedera et al., 2018; Alessi et al., 2019). Trajectories in this class pair should be interpreted together with the continuous LTC layers and the ensemble standard deviation rather than from the argmax class alone. Beech errors are distributed evenly across the four remaining deciduous classes without a dominant confusion pair. The most extreme sink behaviour is shown by "other deciduous", which additionally gets more plots from other classes than its dataset size, which is consistent with its low precision (Tab. 3).

Within the coniferous leaf type classes predictions are more stable. Spruce reaches a recall of 0.81, and all 137 of its misclassified plots remain within the coniferous group (84 silver fir, 35 other coniferous, 18 pine) with the highest precision of 0.96 of any class. Silver fir attains the highest recall (0.89, with 4 % loss to pine and 4 % to beech) and pine reaches 0.70. Conifer confusion is therefore predominantly a problem between structurally and phenologically similar evergreen needle-leaved stands. The CFI matrix is computed on photo-interpreted CFI2020 reference polygons under the direct estimator. Mixed stands remain a challenge and are getting confused more frequently (Blickensdörfer et al., 2024).

The lack of observation density during the Landsat 4/5 period (see Fig. 3) results in misclassifications especially in cloud affected regions such as the Alps for this time period. In early years of the prediction period a significant amount of evergreen oak is classified as dominant species in areas above 1500m and 44°N, which is ecologically implausible. In later periods the models self-correct these pixels to coniferous species. These misclassified evergreen oak pixels have a median level-of-agreement among the model ensembles of 0.58, which is considerably lower than the overall median of 0.84 for the whole dataset. 88.7% of these pixels are below the 25th percentile of the whole dataset's level-of-agreement (0.7). For follow-up studies based on the dataset, the uncertainty measures in both DTS and LTC datasts can be valuable additional information to discard misclassifications and obtain more reliable analysis.

3.1.3 Historical orthophoto imagery

Several sites with orthophoto pairs separated by 12 to 18 years were compared against the annual LTC trajectories and against the difference between the mapped 2020–2025 and 2000–2005 cover means (Fig. B1).At the two sites with progressive conifer succession on previously open land (id 1 and id 3), the mapped CON fraction averaged over the area rises from about 13 to 39 $\%_{cover}$ and from about 16 to 45 $\%_{cover}$ between 1987 and 2025, with a simultaneous decline of DEC fraction. At the site with an abrupt removal (id 6), a dense conifer stand present in the 2008 orthophoto has been cleared in the 2023 image; the mapped CON fraction is stable near 38 $\%_{cover}$ until 2010 and drops to about 15 $\%_{cover}$ within roughly three years, so the event can be dated to the early 2010s from the time series alone, while the orthophotos only bracket it to a 15-year interval.

The dating precision of about three years corresponds to the three-year observation pooling window (see Sec. 2.3). Both the direction and the approximated time of canopy change are reproduced by the annual predictions.

The comparison also exposes the main limitation of the LTC product. Cover fractions are predicted for every pixel inside the ESA WorldCover tree cover mask, irrespective of how much of the pixel is actually covered by trees (Sec. 2.2.6). Where the mask includes grassland, sparsely forested land or recent clear-cuts, the model still predicts values, and the herbaceous and shrub signal is assigned predominantly to DEC, whose seasonal spectral signal is similar. This is visible at id 1, where a DEC fraction above $65\,\%_{cover}$ is mapped for a period in which the orthophoto shows a treeless surface, and at id 4, where an increase of DEC at the expense of CON is occuring within a mixed landsacape of grasslands forest and clearings.

3.1.4 Comparison with published products and scope of the reported metrics

The LTC accuracies are broadly comparable to published fractional composition products, although the comparison is only an indicator. Direct prediction of tree species proportions from Sentinel-2 with a spectral–spatial CNN in Belgium achieved an adjusted R^2 of 0.50 for the predicted proportions and an overall accuracy of 0.73 for the majority species (Bolyn et al., 2022), and the pan-European deciduous–coniferous proportion product derived from Sentinel-2 and harmonised national forest inventory plots reports R^2 values of 0.52–0.73 with relative RMSE between 30 and $65\,\%$ across processing areas (Miettinen et al., 2025). The DEC results reported here ($R^2 = 0.84$) fall above and the EVE results ($R^2 = 0.66$) within that range. But several differences have to be pointed out for the comparison. The reference quantity differs (projected cover per leaf type in this study versus basal-area or biomass proportions in the cited products), the sensor and grid differ ($30\,\mathrm{m}$ Landsat versus $10\,\mathrm{m}$ Sentinel-2), the class definitions are coarser here, and the evaluation samples differ in spatial design.

3.2 Leaf type cover dynamics

Fig. 6b) reproduces the 2025 expected national distribution of the leaf types. CON cover dominates the alpine areas and high northern Apeninnes, DEC cover spreads across the southern alpine regions the whole Apennine chain, and EVE cover dominates in coastal areas and insular Sardegna. Large parts of the Alps are mapped as mixed CON–DEC forest rather than as pure leaf types (Mattioli et al., 2025). To investigate 40 years of change in LTC, a Theil–Sen regression slope (Virtanen et al., 2020) was computed per pixel for the three target classes EVE, DEC, and CON over the annual predictions from 1987 to 2025 (Fig. 7). Additionally, the confidence interval of the slope is calculated to rule out pixels with no significant trend. The trends are only evaluated inside the ESA WorldCover map and describe the changes within the forested pixel rather than the change in forest area (Gasparini et al., 2022).

Across the mapped forest area, $84.5\,\%$ of the pixels have a slope that is outside the confidence area. The overall average rates are small with $-0.9\,\%$ for EVE, $+0.8\,\%$ for DEC, and $+0.1\,\%$ for CON over the whole time period. But the changes are directional for the different leaf types. EVE decreases on $31.8\,\%$ of the forest area and increases on $19.9\,\%$, DEC increases on $30.4\,\%$ against $24.2\,\%$ decreasing, and CON is almost balanced. There is a strong latitudinal contrast in the slope signal. South of $41.5\,^{\circ}\mathrm{N}$, DEC gains $+4.5\,\%_{cover}$ over the record and is the confident trend on $48.6\,\%$ of pixels against $13.9\,\%$ losing, while

CON loses -1.2 and EVE $-1.1\%_{cover}$. North of $44°$N it reverses, with CON gaining $+0.7$ and DEC losing $-0.8\%_{cover}$. Central Italy carries the strongest EVE decline ($-2.0\%_{cover}$, confident on 38.4% of pixels).

The three cover fractions are describing the same pixel, which means the trends are coupled through the leaf type composition. Spatially this means, that 56.7% of the pixels with a significant CON decrease also show a DEC increase, increasing to 63.3% in the south, while 64.4% of the confident CON increases in the north come with a confident DEC decrease. CON losses in the south are concentrated in the Nebrodi range, the Sila massif and Cilento Apennines, where Mediterranean pine stands are concentrated and fire, drought, and deseases disturb pine stands (Nicoletti et al., 2024; Herraiz et al., 2025; Viana-Soto and Senf, 2025). These are also areas with strong DEC gain. CON gain is distributed across the Eastern Alps, which can be related to forest densification and upslope expansion (Leonelli et al., 2011; Tong et al., 2023). EVE gains are most prevalent within the evergreen ranges of Basilicata and Apulia and the Po valley in Northern Italy.

These change rates range below the model performance accoracy of individual pixels. An accumulated slope of $3.8\%_{cover}$ over the whole time period is far below the DEC and EVE MAE values (Tab. 2). Only aggregated patterns can be interpreted, the values reported here are per-pixel fractions of the mapped grid. The Italy-wide EVE decline is in line with recent regional findings (Seddaiu et al., 2025), but seems contradictory to the findings of Conedera et al. (2018) and Alessi et al. (2019), who report evergreen broad-leaved expansion in Italy and the southern Alps. But those studies describe understory laurophyll and evergreen shrub cover and modelled habitat suitability. In our study the annual EVE maps mostly come from a top-of-canopy Landsat signal dominated by *Q. ilex* and *Q. suber*. EVE understory contributes little to the overall Landsat signal under deciduous overstory outside the leaf-off-season (Chastain and Townsend, 2007; Yang et al., 2023). But the error is methodological, as EVE cover and evergreen classes are overpredicted during the sparse observation Landsat-4/5 period (Sec. 3.1) and the positive bias towards higher EVE cover at the start of the time series produces a negative slope, that is most prevalent in the alpine regions. The EVE maps should therefore be used in combination with the confidence layers, trajectories can be estimated for shorter time periods and analysis can be carried out on a regional basis, where forest composition is well understood.

3.3 Dominant tree species dynamics

Climate change induced forest habitat changes manifest geographically as shifts in species distribution across a latitudinal or elevational gradient. We investigate the redistribution of DTS by comparing 1987-1996 and 2016-2025 stable DTS per pixel. The spatial patterns of the tree species during the two time periods are explored as trajectories in latitude and elevation.

The decadal majority classification for 2016–2025 (Fig. 6a)) reproduces established zone distribution of dominant species. Spruce, silver fir and pine dominate in the alpine areas, the Apennine belt is dominated by deciduous oak, beech, and chestnut, and evergreen oak and other evergreens dominating on Sardegna, along the western Italian coastline and the Calabrian peninsula. 89% of the mapped pixels remain stable and carry the same dominant species in the two decadal periods, when restricting to pixels whose ensemble level-of-agreement is at least 70% in both decades. The share of pixels with a level-of-agreement above 70% increases between the two time periods from 74 to 77%, which is consistent with the increase in observation density and therefore more stable predictions (Sec. 2.5).

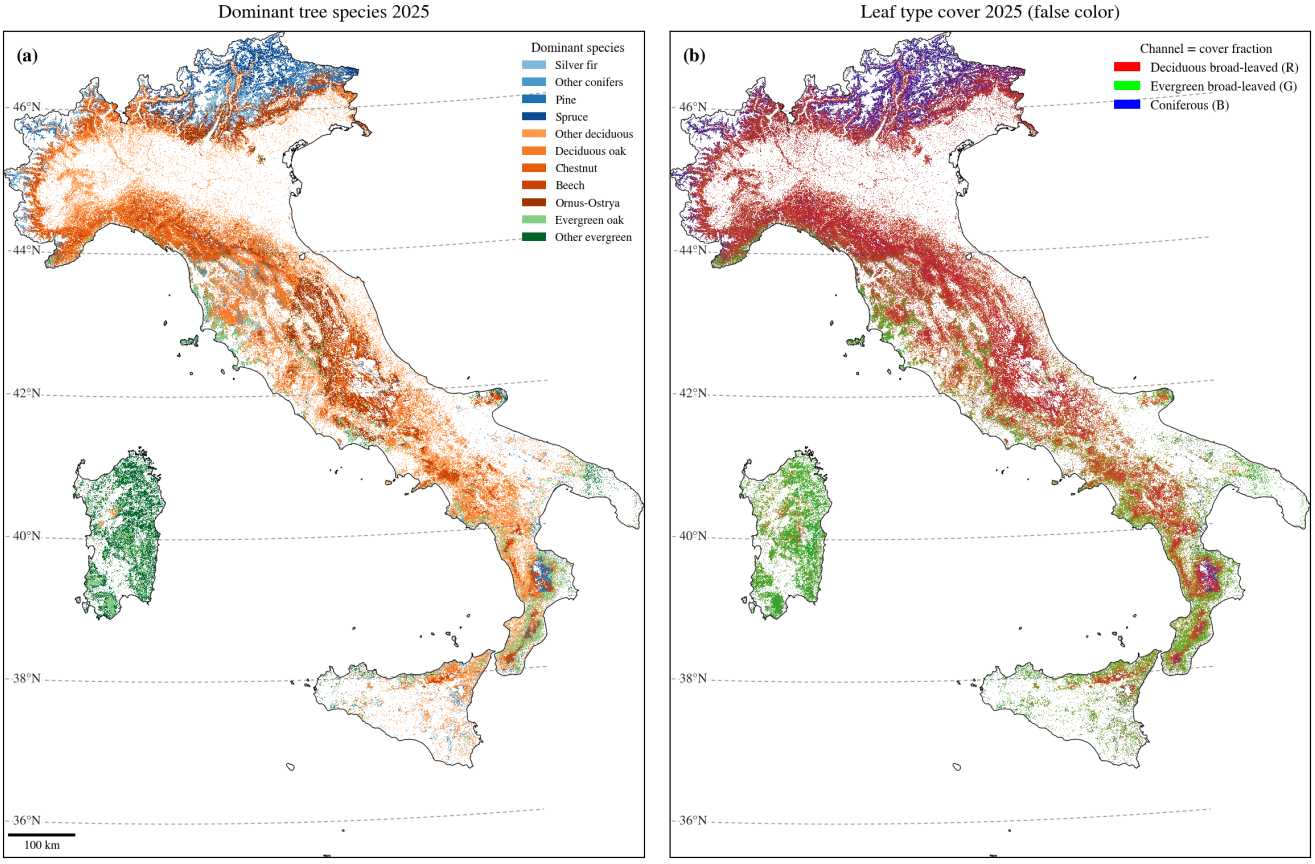


Figure 6. a) Dominant tree species for 2025, which is the last mapped year in the hindcasted annual time series. b) Leaf type cover for 2025 as a False color image mixing the three leaf types per pixel.

Five of the eleven classes show a significant displacement with a p-value < 0.05 of their latitude–elevation distribution between the two decades (Fig. 8). Pine carries the largest drift, with the centroid moving $+0.83^\circ$ of latitude ($90\,\mathrm{km}$) and $+190\,\mathrm{m}$ in elevation, driven by the decrease in the near sea level southern areas present in 1987–1996 and by an extension of the Alpine areas to about $2150\,\mathrm{m}$. Evergreen oak follows the direction expected under warming, moving $+0.24^\circ$ north and $+31\,\mathrm{m}$ upslope, its 2016–2025 distribution reaching both further north and slightly higher. Deciduous oak shifts north by about 0.1° with unchanged elevation and gains areas in the southern Pre-Alps. Chestnut is the only class, that seems like moving south by -0.09°, while gaining $+19\,\mathrm{m}$ ($p = 0.001$), losing its upper margin between 44 and 45°N and gaining low-elevation area in the south. Spruce drift is statistically significant, but its centroid is relatively stationary, the change being an extension of their upper envelopes in their alpine habitats from about 2600 to $2800\,\mathrm{m}$. The remaining six classes show no significant drift.

The directions are generally consistent with the thermophilization reported in Italian forests, where warmer temperatures and increased drought frequency push montane species upslope and favour more drought-resistent species (Leonelli et al., 2011; Chelli et al., 2017; Herraiz et al., 2025). The elevational range of shifts are in a similar magnitude as reported from

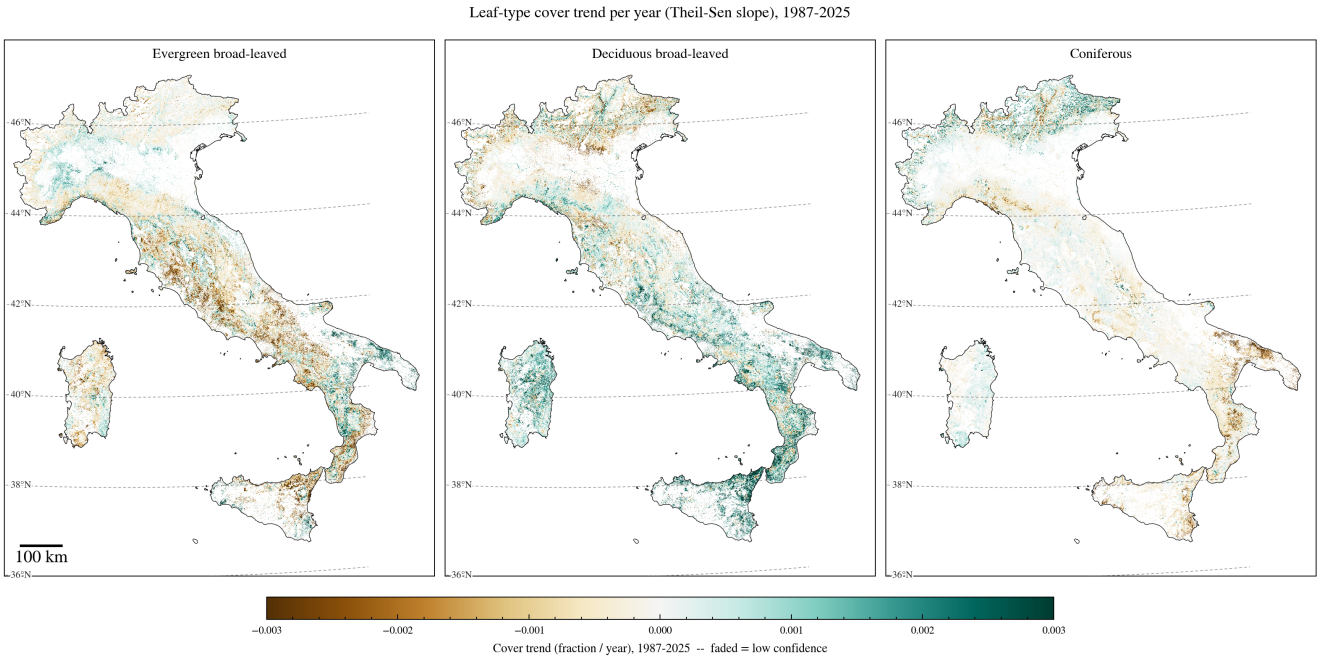


Figure 7. Theil-Sen regression slope of the 3 target LTC classes for years 1987–2025, depicting the annual rate of change for the respective cover class.

vegetation surveys (Midolo et al., 2026). However, the expansion of evergreen broad-leaved species into deciduous stands in the southern Alps has been attributed primarily to land-use changes rather than warming (Conedera et al., 2018). The drifts should not be read as migration of stands, but as drift of the class area, where the direction meets the climatic expectation of the drift. The large pine drift might be inflated by the overprediction of evergreen classes in the low observation desity Landsat-4/5 period (Sec. 3.1) and by the later improved separation of pine from other coniferous classes coming higher observation density. Hence, the $90\,\mathrm{km}$ range shift is an optimistic upper bound. Class confusion happens especially within leaf types (Fig. 5), which makes the drift signal more robust when aggregating to leaf type than operating at species level. The deciduous oak–evergreen oak classes has the largest across-leaf-type confusion and is at the same time the pair of high ecological interest, so it should be interpreted in combination with the continuous LTC layer and the ensemble standard deviation/level-of-agreement. Confirmation of these drift trajectories would require resurveyed plots and a continuous monitoring of stands (Kapfer et al., 2017; D'Amico et al., 2025).

The two products come from two different models. The tension between expanding classes and declining cover fractions highlights the independence of the models, as a pixel can change its dominant species without a matching change in the cover fraction. The agreement between the two mapped products can be a usable consistency check for downstream tasks.

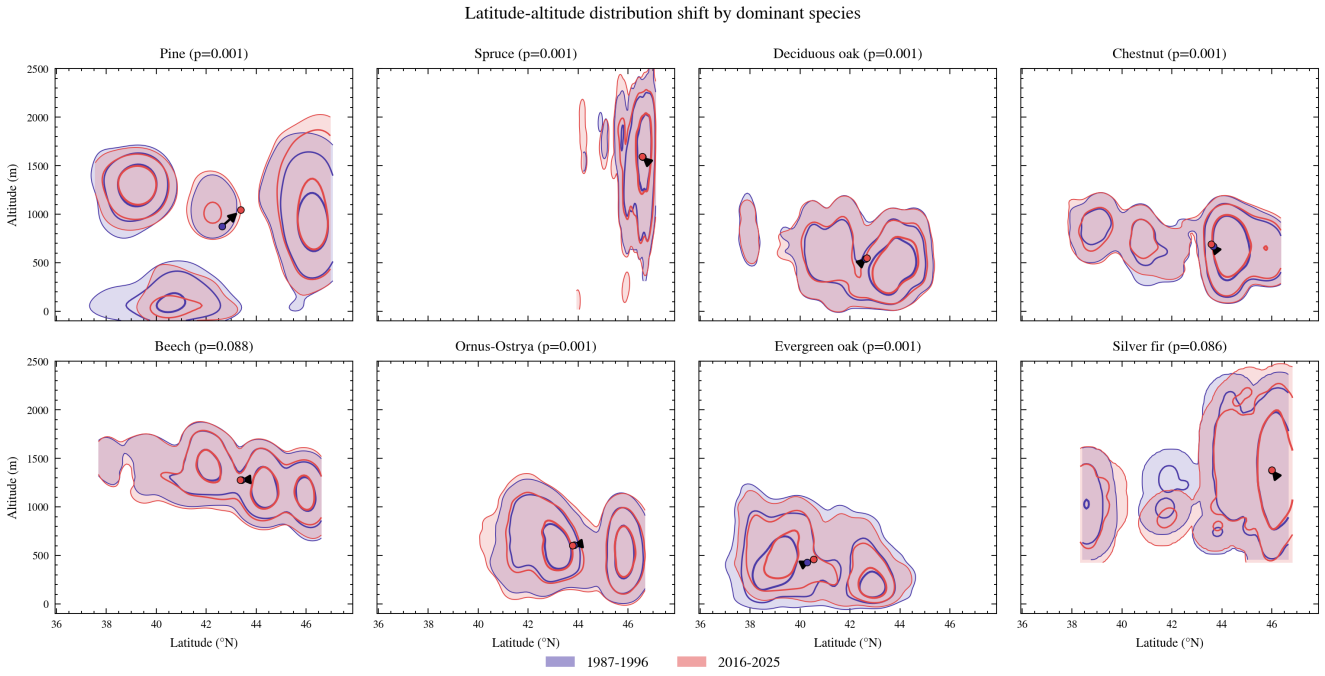


Figure 8. Latitudinal and elevational distribution of DTS between the time periods of 1987–1996 and 2016–2025 for the classes with specific tree species. The arrows indicate the direction and magnitude of the centroid shift between the two time periods.

3.4 Limitations and future work

The reported accuracy of this dataset is limited more by the uncertainty of the reference data than by the model predictions. Reported VDBI location accuracies span two orders of magnitude (Sec. 2.2.1), and a coordinate displaced by more than one pixel attaches its label to an unrelated stand, degrading both training and the reported metrics. Receivers require long occupations to reach metre-level accuracy under closed canopy (Strunk et al., 2025). Such reference error biases accuracy assessments negatively (Elmes et al., 2020; Persson et al., 2022), which is why the VDBI figures in Tab. 2 are bracketed by three estimators rather than reported as a single value. Although the brackets can not be seen as a correction: likelihood-based geolocation correction is unbiased only against an independent high-accuracy reference (Shannon et al., 2024), which does not exist for the historical period. The VDBI-based figures should therefore be read as bounds and the VPO-based figures as the operationally relevant accuracy.

No dataset available for Italy allows the mapped trajectories to be validated directly. VDBI is the only reference source distributed across the mapped time period, and its plots are clustered in well-studied vegetation types and regions, a bias common to European vegetation-plot archives (Chytrý et al., 2016; Sabatini et al., 2021). Reported accuracies therefore describe per-pixel performance at plot locations and are not area-adjusted map accuracies (Olofsson et al., 2014). Users requiring the latter for a specific region or period should conduct a probability-based assessment on the published maps. Resurveys of historical

vegetation plots (Kapfer et al., 2017; Verheyen et al., 2017; Midolo et al., 2026) and the reorientation of the Italian national inventory towards continuous monitoring (D'Amico et al., 2025) are the two necessary steps to a full time trajectory validation in future studies.

Annual predictions are produced independently, with no temporal model linking individual years. Three-year pooling smoothes, but does not remove inter-annual variability caused by decreasing observation density, cloud contamination or inter-annual phenological differences, so DTS pixels may move between spectrally similar species in consecutive years even when the stand is actually stable. We publish unsmoothed and unprocessed time series predictions (Sec. 2.5), but users are encouraged to apply andvanced methods to stabilize the time trajectories. For instance, a hidden Markov formulation with per-year ensemble probabilities as emissions, an ecologically constrained transition matrix, and Viterbi smoothing, could suppress such flips while preserving true class transitions (Abercrombie and Friedl, 2016; Hermosilla et al., 2018), especially if combined with spectral trajectories (Kennedy et al., 2010; Zhu and Woodcock, 2014) or independently mapped disturbance dates (Viana-Soto and Senf, 2025). This would be a promising improvement to the DTS layer, which could also be applied to the LTC layer in an adapted way.

The input archive itself remains heterogeneous. Usable satellite observations per pixel and year increase from the Landsat 5 period to present (White et al., 2014; Bayle et al., 2024), SLC-off gaps in Landsat-7 observations affect all years from 2003 onwards, and radiometric differences between the different sensors aboard the Landsat missions (TM, ETM+ and OLI) persist after Level-2 processing (Roy et al., 2016). The sensor embedding and three-year pooling make the model aware of these effects but cannot separate true vegetation change from artefacts of changing observation conditions. A cross-calibrated, gridded full-archive product in the sense of Landsat Analysis Ready Data (Dwyer et al., 2018), FORCE (Frantz, 2019) or the Harmonized-Landsat-Sentinel-2 archive (Claverie et al., 2018) would remove much of this uncertainty. In the current case the pre-1999 maps have the highest uncertainty of the record and all maps should be used together with the ensemble standard deviation layers.With the Landsat archive spanning over four decades and reaching a time scale relevant to ecological change, more research is needed to harmonize the different Landsat sensors and make large-scale time series analysis more accessible.

Mapping cost in terms of compute time is a practical constraint on how often the product can be updated. Although parallelized, fetching, masking and featurising three years of observations per target year for every forested pixel and running five ensemble members required several weeks of continuous processing, dominated by data transfer and preprocessing, and finally model inference (Gorelick et al., 2017; Frantz, 2019). Pretrained representations are the most promising route out of both this and the label constraint, since analysis-ready embeddings reduce a year of observations to a compact per-pixel vector consumable by lightweight models (Brown et al., 2025; Feng et al., 2026; Ball et al., 2026; Hiebl et al., 2026b). The synthetic LTC labels used here were generated this way (Sec. 2.2.4). Existing embedding products, however, cover only the modern multi-sensor era ($\approx 2015 - \text{present}$). A Landsat-native backbone pretrained on unlabelled pixel time series (Tseng et al., 2024; Stewart et al., 2025) would extend this benefit to the full available Landsat archive and reduce each new mapping task to a fine-tuning step. Such a foundation model or dataset would also address the previously mentioned inter-sensor calibration issues.

4 Conclusions

The Landsat collections provide an unprecedented archive of satellite imagery for reconstructing and hindcasting environmental variables across large regions. Few studies exist, that exploit the full 40 years of available Collection-2 imagery to map forest related variables and provide time series datasets, that can be used to investigate forest dynamics and changes under climate and land-use change conditions. This study predicts annual maps of dominant tree species and leaf type cover from Landsat time series, forest plot observations and forest inventories. The resulting datasets cover every forested pixel of Italy at $30\,\mathrm{m}$ resolution for each year from 1987 to 2025, and reports a five-seed ensemble standard deviation and, for the species layer, a level-of-agreement alongside every prediction. On recent field plots the cover fractions reach an MAE of $9.7\,\%_{cover}$ for DEC ($R^2 = 0.84$), $12.5\,\%_{cover}$ for EVE ($R^2 = 0.66$) and $5.0\,\%_{cover}$ for CON, and dominant species are recovered with an overall accuracy of 0.62 to 0.71 depending on the test dataset.

The mapped distributions reproduce established forest zonations, with CON dominating the Alps and northern Apennines, DEC spanning the Apennine chain and southern Alps, and EVE dominating coastal areas and Sardegna, with $89\,\%$ of pixels preserving the same dominant species between the 1987–1996 and 2016–2025 decades. Within this stability, both products shows directional changes that emerge when aggregated. DEC cover gains over EVE and CON losses in the south and a significant latitude–elevation drift is detected for pine, evergreen oak, deciduous oak and spruce, consistent with thermophilization processes. The trends are generally more reliable when read at leaf-type and spatially or temporally aggregated scale, filtered by the uncertainty layers. Factors that especially limit the capability of the product, are the geolocation error of the plot observations, and the sparse-observation Landsat 4/5 period that carries the highest uncertainty of the record. Closing these gaps and addressing the challenges in long-term forest mapping on a large scale require resurveyed plots and continuous forest-inventory monitoring to validate the mapped trajectories, a cross-calibrated, harmonized full-archive Landsat product to remove inter-sensor variability, or a Landsat-native pretrained backbone to extend embedding-based mapping across the full record and lower the barrier for future mapping tasks.

5 Code and data availability

The datasets, model weights and training data are available on https://doi.org/10.48323/4v6da-v5k69 (Hiebl et al., 2026a). The full LTC dataset (cover and standard deviation) can be downloaded as geotiff files. As the DTS dataset including majority class, level-of-agreement, probabilities, and probability standard deviations exceeds several $100\,\mathrm{GB}$ in size, we only make class predictions and level-of-agreement downloadable via the above mentioned repository. Probabilities and standard deviations per class are available as zarr-compressed files upon request at the authors.

All python code that was used during data processing, training and mapping is available under https://github.com/benehiebl/ls_italy_forest (training) and https://git.uibk.ac.at/rslab/sattstools (beta version, remote sensing data processing).

Table A1. Overview of the three estimators used to evaluate the mapped results.

	Definition	Use of reference label	Characteristic
direct	map value at the pixel containing the reported plot coordinate, in the reported observation year	no	pessimistic lower bound, which attributes all reference related geolocation error to the model
posterior-weighted	mean over the space–time neighborhood of the reported location accuracy, weighted by Gaussian location/time prior times a Gaussian reference label likelihood	yes	location accuracy based estimate
best-match	reports the single candidate, that agrees best with the reference label within the reported location accuracy	yes	optimistic upper bound

Appendix A: Computation of evaluation metrics

All accuracy figures reported in Sec. 3.1 are computed from the published maps, this means the estimators below are applied to the same spatio-temporal product that is published, and not to model outputs held in memory. For a plot p with reported coordinates $\mathbf{s}_p$ and observation year t_p, let $\mathcal{N}_p$ be the set of candidate map pixels $(\mathbf{s}, t)$ with $\|\mathbf{s}-\mathbf{s}_p\|_\infty \leq 2\sigma_s$ and $|t-t_p| \leq 2\sigma_t$, clipped to the temporal extent of the product. This creates a squared box of half-width $2\sigma_s$ around the plot coordinates. The *direct* estimator is the special case $\mathcal{N}_p = \{(\mathbf{s}_p, t_p)\}$, i.e. the nearest pixel of the reported year. The two estimators described below are conditioned on the reference plots label and are therefore validation diagnostics only, as they cannot be applied where the reference value is unknown. Pixels without a valid prediction receive zero weight and are excluded, and plot–prediction pairs containing a missing value are dropped before metrics are computed. The three estimators characteristics are summarized in Tab. A1.

A1 Posterior-weighted estimator

Every candidate pixel in $\mathcal{N}_p$ is treated as a hypothesis for the true plot location and observation date. The prior expresses the reported position and year and their uncertainty as a Gaussian kernel,

$$w^{\text{prior}}(\mathbf{s}, t) = \exp\left(-\frac{\|\mathbf{s}-\mathbf{s}_p\|^2}{2\sigma_s^2}\right)\exp\left(-\frac{(t-t_p)^2}{2\sigma_t^2}\right), \tag{A1}$$

with position uncertainty $\sigma_s =$ plot location accuracy in m and temporal uncertainty $\sigma_t = 1\,\mathrm{a}$, clamped at $2\sigma_s$ and $2\sigma_t$. For the LTC task the likelihood of a hypothesis is the agreement between the mapped cover fraction $\hat{y}_k(\mathbf{s}, t)$ of leaf type $k \in$

$\{\mathrm{EVE}, \mathrm{DEC}, \mathrm{CON}\}$ and the plot reference cover $y_{p,k}$,

$$w_k^{\mathrm{lik}}(\mathbf{s}, t) = \exp\left(-\frac{\left(\hat{y}_k(\mathbf{s}, t) - y_{p,k}\right)^2}{2\sigma_{\mathrm{obs},k}^2}\right), \tag{A2}$$

and the estimate is the posterior-weighted mean over the neighbourhood,

$$\hat{y}_{p,k}^{\mathrm{PW}} = \frac{\sum_{(\mathbf{s},t)\in\mathcal{N}_p} w^{\mathrm{prior}} w_k^{\mathrm{lik}} \hat{y}_k(\mathbf{s}, t)}{\sum_{(\mathbf{s},t)\in\mathcal{N}_p} w^{\mathrm{prior}} w_k^{\mathrm{lik}}}. \tag{A3}$$

The observation bandwidth $\sigma_{\mathrm{obs},k}$ is set per leaf type to the standard deviation of the residuals of the direct estimator on the VPO split, which is the dataset with the lowest location uncertainty, so that the likelihood is not sharper than the residual scatter that is already explained by the model error. It acts as a trade-off between the two limiting cases: $\sigma_{\mathrm{obs},k} \to \infty$ removes the label from the weighting and recovers a purely prior-weighted neighbourhood mean, whereas $\sigma_{\mathrm{obs},k} \to 0$ recovers the best-match estimator of Sec. A2. This is similar to a sampling-importance-resampling implementation that draws n candidates directly from the prior and weights them by the likelihood (Lobo et al., 2020).

For the DTS task the same weights are applied to the predicted class-probability vector $\hat{\mathbf{p}}(\mathbf{s}, t)$. The likelihood of the hypothesis is the probability that the candidate cell assigns to the reference class.

Spatial uncertainty σ_s is set for each plot individually, depicting the reported accuracy of the plot coordinates based on GNSS measurements or data curator expert knowledge. σ_s ranges between $1\,\mathrm{m}$ and $1000\,\mathrm{m}$ in the VDBI dataset, which guides the prior distribution over the spatial dimension and the size of the neighbourhood (see Fig. A2). The sensitivity of the posterior-weighted estimator to the choice of $\sigma_{\mathrm{obs},k}$ is illustrated in Fig. A1, and example posterior landscapes are shown in Fig. A2.

A2 Best-match estimator

The best-match estimator replaces the weighted mean by a hard selection over a neighbourhood ($2\sigma_s$, plot location accuracy in m, and a fixed window of $t_p \pm 1\,\mathrm{a}$). For the LTC task it returns, per leaf type, the value of the single candidate cell that agrees most closely with the reference,

$$\hat{y}_{p,k}^{\mathrm{BM}} = \hat{y}_k\left(\underset{(\mathbf{s},t)\in\mathcal{N}_p}{\arg\min} \left|\hat{y}_k(\mathbf{s}, t) - y_{p,k}\right|\right), \tag{A4}$$

and for the DTS task the label of the candidate cell that assigns the highest probability to the reference class. Because the selection is made using the reference value itself, this estimator is optimistically biased by itself: any leaf-type value present anywhere in the neighbourhood can be matched, and a plot is counted as correctly classified whenever at least one candidate cell within the plot location accuracy and $\pm 1\,\mathrm{a}$ carries the reference class. The resulting metrics are therefore not corrected accuracies but an upper optimistic bound on what could be attributed to spatial(-temporal) reference mismatch rather than to model error.

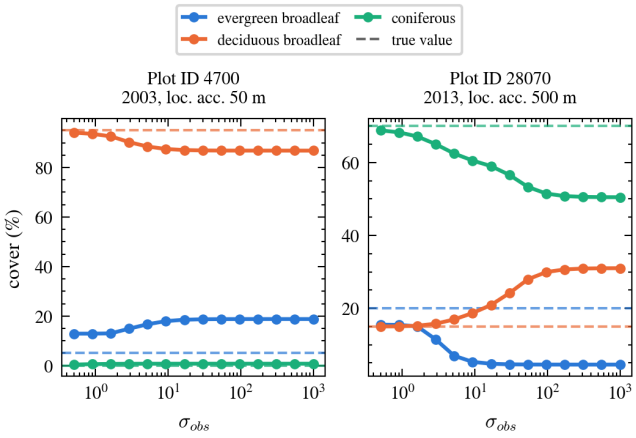


Figure A1. Sensitivity study of the observation bandwidth $\sigma_{\mathrm{obs},k}$ that controls the reference-based likelihood landscape for the posterior-weighted estimator. With lower $\sigma_{\mathrm{obs},k}$ the LTC estimate approaches the best-match and therefore true observation value (dashed lines). Plot 28070 inherits a high uncertainty as its location uncertainty is high and the neighborhood pixels are heterogenous. While Plot 4700 has low location uncertainty where best-match and posterior weighted estimates are relatively similar.

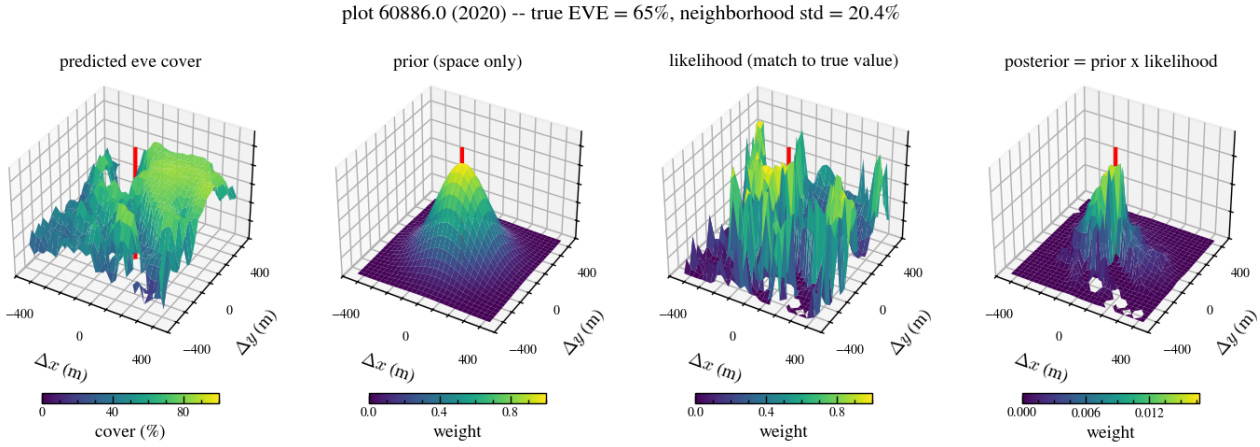


Figure A2. Posterior weight landscape for an example plot with highly heterogenous EVE cover within the neighbourhood computed from the gaussian prior and likelihood for the posterior-weighted estimator. The LTC estimate is weighted within the obsevation window defined by the prior distribution and the likelihood of the reference value, downweighting candidate pixels that are further away and specifically dissimilar to the reference value.

Appendix B: Comparison to historical orthophoto imagery

Fig. B1 shows several examples of forest patches that visibly changed over the last decades by various reasons. For further details see Sec. 3.1.3

Author contributions.

B.H.: Conceptualization, Writing - original draft, Methodology, Software, Formal Analysis, Validation, Data curation. N.A.: Writing - review & editing, Data curation. G.B.: Writing - review & editing, Data curation. G.C.: Data curation. A.B.: Writing - review & editing. S.C.: Writing - review & editing, Visualization, Validation. G.Z.: Writing - review & editing. S.Z.: Writing - review & editing, Project administration. M.R.: Writing - review & editing, Resources, Project administration.

Competing interests.

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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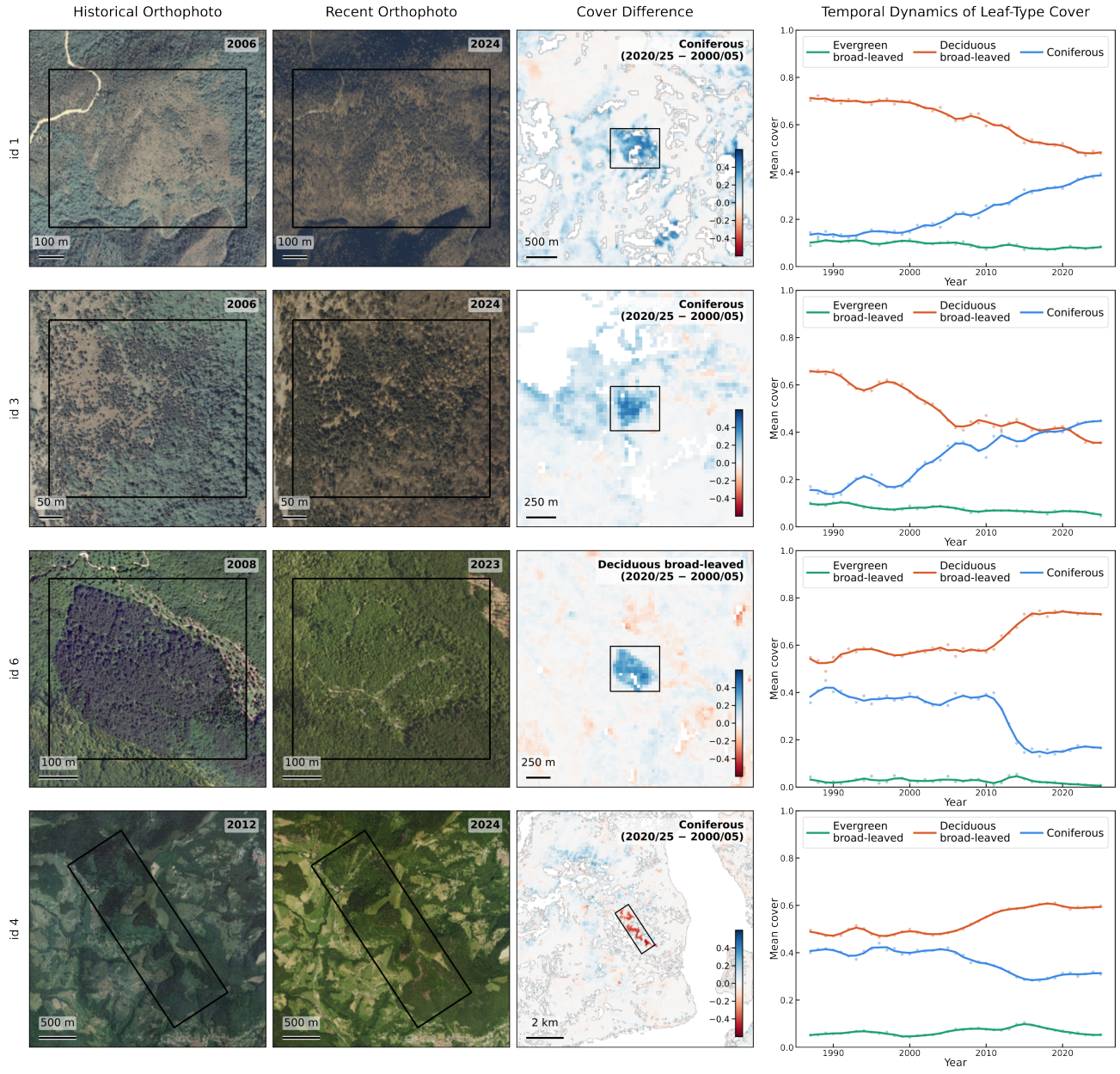


Figure B1. Comparison of four sites (rows) between historical and recent orthophotos (first two columns), the difference between the mapped 2020–2025 and 2000–2005 mean cover of the indicated leaf type (third column), and the annual LTC trajectories averaged over the delineated area (fourth column). Sites id 1 and id 3 show progressive coniferous encroachment on formerly open land, reproduced by the maps as a gradual CON increase from about 2000 onwards at the expense of DEC. Site id 6 shows a coniferous stand cleared between the 2008 and 2023 orthophotos, which appears in the time series as an abrupt CON drop in the early 2010s taken over by the deciduous signal. Sites id 1 (before the encroachment) and id 4 illustrate the limitation of the forest mask: grassland, clear-cuts and sparsely treed pixels are also mapped, and their herbaceous signal is assigned predominantly to DEC.